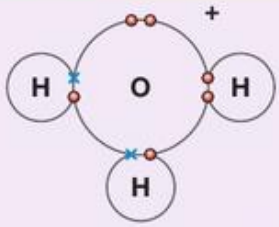
HINT

You do not have to remember the names of the theories.

EXTENSION WORK



In the dot-and-cross diagram for the H_3O^+ ion you can see that the covalent bond that formed between the H_2O and the H^+ is unusual in that both the shared electrons come from the O. This is called a dative covalent bond or a co-ordinate covalent bond. Once it has been formed, it is exactly the same as a 'normal' covalent bond.

KEY POINT

You have already met the ammonium (NH_4^+) ion, which is obtained when H^+ is added to ammonia (NH_3). The hydroxonium ion is the equivalent ion for water. You do not have to remember the name *hydroxonium ion*.

KEY POINT

To understand the reaction between ammonia and hydrochloric acid in relation to the Arrhenius theory we have to realise that ammonia reacts with water to form OH^- ions:

$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$

and that hydrochloric acid dissociates to form H^+ ions:

$HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$

and that the overall reaction involves H^+ ions reacting with OH^- ions to form water:

$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$

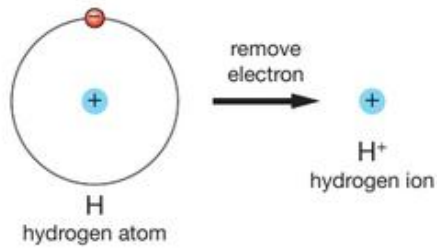


▲ Figure 17.17 Ammonia and hydrogen chloride gases also react to form ammonium chloride (the white smoke): $NH_3(g) + HCl(g) \rightleftharpoons NH_4^+Cl^-(s)$

There is a more general theory of acids and bases that can be used in more situations:

- *An acid is a proton (hydrogen ion) donor.*
- *A base is a proton (hydrogen ion) acceptor.*

This is called the Brønsted–Lowry theory.

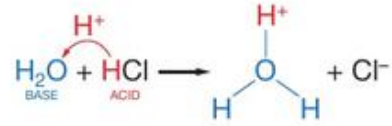


▲ Figure 17.14 A hydrogen ion (H^+) is a proton. If an electron is removed from a hydrogen atom, only a proton is left.

A better way of writing the reaction that occurs when hydrogen chloride dissolves in water is:

$$H_2O(l) + HCl(aq) \rightarrow H_3O^+(aq) + Cl^-(aq)$$

The HCl donates (gives) a proton (H^+) to an H_2O molecule. So, according to our definition, the HCl is an acid and H_2O is a base because it accepts the proton.



▲ Figure 17.15 HCl donates a proton and H_2O accepts it.

The $H_3O^+(aq)$ ion is called the **hydroxonium ion**. This is the ion that we usually write simply as $H^+(aq)$. You can think of it as a hydrogen ion joined to a water molecule. Cl^- has a negative charge because the H^+ left an electron behind when it transferred to the H_2O.

THE REACTION BETWEEN NH_3 AND HCl

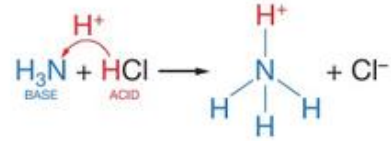
Dilute hydrochloric acid reacts with ammonia solution in a neutralisation reaction to form a salt, ammonium chloride:

$$NH_3(aq) + HCl(aq) \rightarrow NH_4Cl(aq)$$

Ammonium chloride is an ionic salt so we can show the ions that are present:

$$NH_3(aq) + HCl(aq) \rightarrow NH_4^+(aq) + Cl^-(aq)$$

We can see now that, in order to form ammonium chloride, the HCl must donate a proton (H^+) to the NH_3. The HCl is an acid because it donates a proton and the NH_3 is a base because it accepts the proton.



▲ Figure 17.16 HCl is an acid and donates a proton to NH_3 (a base).

MORE ACID-BASE REACTIONS

Acids in solution are acidic because they donate a proton (H^+) to water to form the hydroxonium ion. If we take a general acid (HA), this is what happens when it is put into water:

$$HA(aq) + H_2O(l) \rightarrow H_3O^+(aq) + A^-(aq)$$

HA donates a proton to H_2O, which acts as a base. For nitric acid this would be:

$$\underset{\text{acid}}{HNO_3(aq)} + \underset{\text{base}}{H_2O(l)} \rightarrow H_3O^+(aq) + NO_3^-(aq)$$

When bases, such as metal oxides (e.g. Na_2O), react with water they accept a proton from water, for example:

$$(Na^+)_2O^{2-}(s) + H_2O(l) \rightarrow 2Na^+(aq) + 2OH^-(aq)$$

The O^{2-} ion accepts a proton from the water. In this reaction the water acts as an acid because it donates a proton.

When sulfuric acid reacts with copper(II) oxide (CuO) we have:

$$Cu^{2+}O^{2-}(s) + H_2SO_4(aq) \rightarrow Cu^{2+}(aq) + SO_4^{2-}(aq) + H_2O(l)$$

H_2SO_4 is an acid, it donates protons to CuO, the base.

Ammonia is also a base:

$$\underset{\text{base}}{NH_3(aq)} + \underset{\text{acid}}{H_2O(l)} \rightleftharpoons NH_4^+(aq) + OH^-(aq)$$

The ammonia accepts a proton (H^+) from the water.

For reactions such as:

$$HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$$

What actually happens is that the H_3O^+ ion formed when the acid reacts with water donates a proton to the base, OH^-.

$$\underset{\text{acid}}{H_3O^+(aq)} + \underset{\text{base}}{OH^-(aq)} \rightarrow 2H_2O(l)$$

But for International GCSE purposes, we almost always use the simplified version:

$$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$$

HINT

Water can act as proton acceptor (base) or proton donor (acid) in reactions on this page. Do not get confused with what is happening here; remember, water is still neutral! We are simply understanding the reactions that occur in terms of proton transfer.

CHAPTER QUESTIONS

SKILLS CRITICAL THINKING 5

SKILLS PROBLEM SOLVING 9

1 State which of the following will react with dilute sulfuric acid:

a copper, copper(II) oxide, copper(II) hydroxide, copper(II) carbonate

b In the case of each of the substances which does react, write the equation (including state symbols) for the reaction. All of these substances are insoluble solids.

SKILLS CRITICAL THINKING 7

2 Read this description of the chemistry of metal **A** and some of its compounds, and then answer the questions.

Metal **A** has no reaction with dilute hydrochloric acid or dilute sulfuric acid. It forms a black oxide, **B**, which reacts with hot dilute sulfuric acid to give a blue solution, **C**. Metal **A** also forms a green compound, **D**, which reacts with dilute nitric acid to give a colourless gas, **E**, and another blue solution, **F**. The colourless gas, **E**, turned limewater milky.

a State the names of **A**, **B**, **C**, **D**, **E** and **F**.

SKILLS PROBLEM SOLVING 9

b Write equations for the reactions between:

i **B** and dilute sulfuric acid

ii **D** and dilute nitric acid.

SKILLS CRITICAL THINKING 6

3 a Nickel, Ni, is a silvery metal just above hydrogen in the reactivity series. Nickel(II) compounds in solution are green.

i Describe what you would see if you warmed some nickel with dilute sulfuric acid in a test-tube.

ii State the name of any gas formed.

SKILLS PROBLEM SOLVING 9

iii Write the equation for the reaction between nickel and dilute sulfuric acid.

b Nickel(II) carbonate is a green, insoluble powder. A spatula measure of nickel(II) carbonate is added to some dilute hydrochloric acid in a test-tube.

SKILLS CRITICAL THINKING 6

i Describe what you would observe in this experiment.

SKILLS PROBLEM SOLVING 9

ii Write an equation for the reaction that occurs.

SKILLS EXTENSTION QUESTION 8

iii Write an ionic equation for the reaction that occurs.

SKILLS CRITICAL THINKING 6

4 Organise the following compounds into two lists: those that are soluble in water and those that are insoluble.

sodium chloride, lead(II) sulfate, zinc nitrate, calcium carbonate, iron(III) sulfate, lead(II) chloride, potassium sulfate, copper(II) carbonate, silver chloride, aluminium nitrate, barium sulfate, ammonium chloride, magnesium nitrate, calcium sulfate, sodium phosphate, nickel(II) carbonate, chromium(III) hydroxide, potassium dichromate(VI)

SKILLS EXECUTIVE FUNCTION

5 a Describe in detail the preparation of a pure, dry sample of copper(II) sulfate crystals, $CuSO_4 \cdot 5H_2O$, starting from copper(II) oxide.

SKILLS PROBLEM SOLVING 9

b Write equations for **i** the reaction producing copper(II) sulfate solution and **ii** the crystallisation reaction.

CHEMISTRY ONLY

SKILLS EXECUTIVE FUNCTION 7

6 a Read the following description of a method for making sodium sulfate crystals, $Na_2SO_4 \cdot 10H_2O$, and then explain the reasons for each of the underlined phrases or sentences.

25 cm^3 of sodium carbonate solution was transferred to a conical flask using a pipette, and a few drops of methyl orange were added. Dilute sulfuric acid was run in from a burette until the solution became orange. The volume of acid added was noted. That same volume of dilute sulfuric acid was added to a fresh 25 cm^3 sample of sodium carbonate solution in a clean flask, but without the methyl orange. The mixture was evaporated until a sample taken on the end of a glass rod crystallised on cooling in the air. The solution was left to cool. The crystals formed were separated from the remaining solution and dried.

SKILLS PROBLEM SOLVING

b Write equations for i the reaction producing sodium sulfate solution and ii the crystallisation reaction.

SKILLS REASONING, PROBLEM SOLVING

7 Suggest solutions that could be mixed together to make each of the following insoluble salts. In each case write the equation for the reaction you choose.

a Silver chloride
b Calcium carbonate
c Lead(II) sulfate
d Lead(II) chloride

SKILLS EXECUTIVE FUNCTION, PROBLEM SOLVING

8 Describe in detail the preparation of a pure, dry sample of barium carbonate. Write the equation for the reaction you use.

9 There are three main methods of making salts:

A Reacting an acid with an excess of a suitable solid.

B Using a titration.

C Using a precipitation reaction.

For each of the following salts, write down the letter of the appropriate method, and state the names of the substances you would react together. You should state whether they are used as solids or solutions. Write an equation for each reaction.

a Zinc sulfate
b Barium sulfate
c Potassium nitrate
d Copper(II) nitrate
e Lead(II) chromate(VI) (a bright yellow insoluble solid; chromate(VI) ions have the formula CrO_4^{2-}).

END OF CHEMISTRY ONLY

SKILLS REASONING

10 All of the following equations represent acid–base reactions. For each of the equations, state which substance is the acid and which the base.

a $CO_3^{2-}(s) + H_3O^+(aq) \rightarrow HCO_3^-(aq) + H_2O(l)$
b $MgO(s) + H_2SO_4(aq) \rightarrow MgSO_4(aq) + H_2O(l)$
c $HNO_3(aq) + NH_3(aq) \rightarrow NH_4^+(aq) + NO_3^-(aq)$
d $H_2SO_4(aq) + H_2O(l) \rightarrow HSO_4^-(aq) + H_3O^+(aq)$
e $NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$
f $CH_3COOH(aq) + NH_3(aq) \rightarrow CH_3COO^-(aq) + NH_4^+(aq)$